MNLP 2026: Homework 1 Presentation

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Definition of attending student

- **Attending students** are students who regularly attend the course. They have to complete **two homeworks**:
  - one to be delivered <u>during the course</u>
  - one to be delivered <u>10 days before each exam session</u> and, in any case, <u>**by the September session**</u>

- **Non-attending students** are students submitting both homeworks at any exam session (including January and February) and/or attending students who failed the homeworks. They must take the **full exam** (more details in the next slides)
  - Their homeworks must be delivered <u>10 days before each exam session</u> (for instance, if the exam date is December 25th, the homework will be delivered by 15th by midnight++ (the ++ means we will not be strict with the submission hour)

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Homework Organization

Each HW has *base requirements* and *additional requirements*:

- **Attending students**: must explore *at least one* additional requirement beyond base requirements for the HW to be accepted
- **Non-attending students** (i.e., *students taking the full exam*) must explore more additional requirements (*at least two*) beyond base requirements for the HW to be accepted

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Homework Organization

Homeworks will be carried out in groups of 2:

- → Form to register your group (**one registration per group only, one of the students represents the group**)
- Use the **representative's institutional mail** (Erasmus students should use their Home University email)

Groups cannot be changed:

- The same group **must do both HW1 and HW2**
- All group members **are bound to present together (same session, same day)** at the final exam, so please make sure that you book those days for your joint presentation **as soon as the exam sessions are published**
- Presentations should be **personal and individual** (i.e., *different*, no slide overlap) for each group member – this means **the same project is presented twice in a row**

If group members want to split, they can do it at any time **BUT**:

- If done *after* the delivery of the first HW, **they will no longer be considered attending students**

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Homework Evaluation

Grading Approach:

- HW1: Binary assessment (Pass/Fail)
  - A “Pass” may include “Low-Pass” cases
- HW2: Binary assessment (Pass/Fail)
  - A “Pass” may include “Low-Pass” cases
- Final grade is the result of the oral exam
  - Keeping into account the quality of HWs and performance during the oral exam

The **oral exam** will revolve around:

- Homework presentations (overall **10 minutes**)
- Starting from your homeworks, theory questions (potentially on the **whole course program**)

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HWs Evaluation: Low-Pass

Categories of **common errors** include (but are not limited to):

- Partial compliance with the **requirements** (e.g. report format, expected work, etc.)
- Low-quality **presentation** (quality of text, layout, clarity, etc.)
- Weak **experimental setup**: superficial comparison/discussion of systems/models/baselines, etc.
- One or more **theoretical errors** (e.g., metrics implemented in wrong ways)
- Difficulty/Problems in **code execution**

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Potential causes of failure

These are strict, indisputable, objective reasons for failure:

- **Missing** report
- **Wrongly formatted** report (e.g., not using the ACL format) – this is unfair to the other students
- **Missing** files (e.g., missing code or mandatory files)
- **Missing** sections in the report
- **Missing mandatory** (base) requirements
- **Lower** number of additional requirements
- **Plagiarism**

Therefore no redressing request will be accepted.

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Important information about your exam

- The exam will be divided into an **oral presentation** and **a series of questions**
- You will have **10 minutes to present your work**. Managing the time of your presentation is part of the skills required to pass the exam
- The questions will **cover both the homework and theoretical knowledge**
- Students will be expected to justify their **approach** to the homework, the **setup** they used, the **metrics**, **baselines** etc.
- Theoretical questions will focus on the foundational knowledge covered in class

Even if your HWs are very good you can **still fail the exam** if your **oral exam** does **not match** the **content** or **quality of your HWs**!

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Homework 1 Evaluation

The 10 top-performing groups in HW 1 will be rewarded with an **exclusive t-shirt** and special **access to the CINECA's Leonardo supercomputer for HW 2**

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Getting help

If you want to reach out for further clarification, you have two options:

1. Write a post on Classroom
   a. First search for relevant or related posts
   b. If you can't find anything, then make a post yourself
2. Send us an e-mail if the issue concerns only your project and nobody else
   a. The object of the e-mail MUST start with "**[MNLP 2026]**"
      i. If you don't use this object, our automatic filters will not find your email → NO REPLY
      ii. Put **everyone** (see below) in CC, so you get more chances of a faster response

Mail: sapienzanlp@diag.uniroma1.it (this will reach all TAs + Prof. Navigli)

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Homework 1 - Retrieval

Semantic Search with Retrieval and Fine-Tuning

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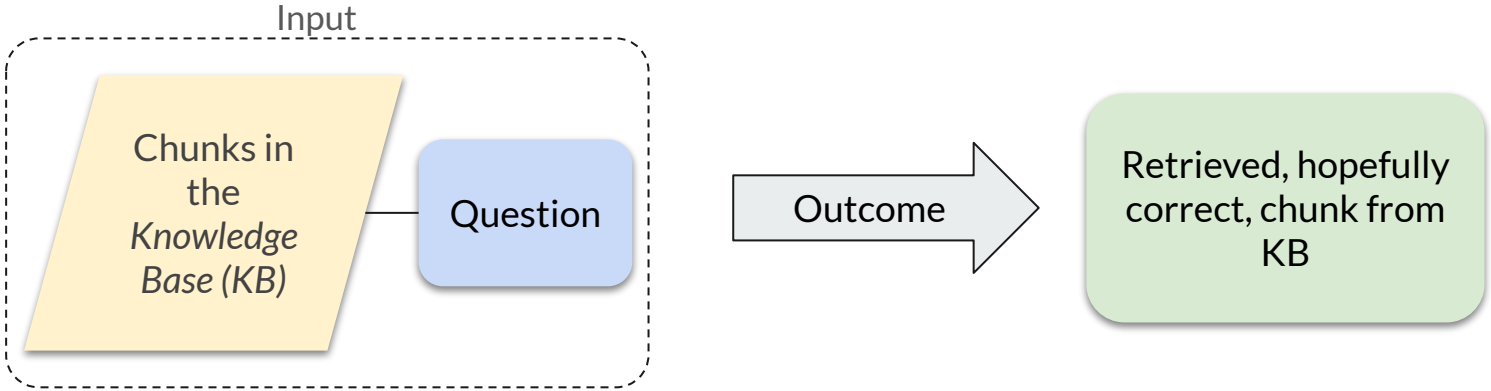
HW 1 - Task Definition

Retrieval via Semantic Search (1/4)

The first homework consists in the creation and analysis of a **semantic search** system (text encoding + retrieval) in a Question-Answering (QA) setting.

You are given a **question** and a set of **chunks**, only one of which is the **right one to retrieve** (i.e. expected to contain the answer).



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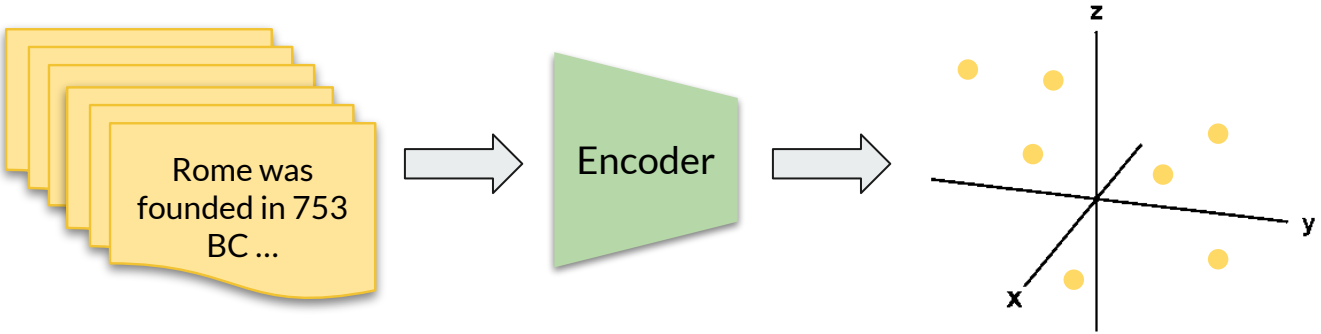
HW 1 - Task Definition

Retrieval via Semantic Search (2/4)

You will be given a single text broken into small chunks.

Goal 1: encode the chunks into fixed-size, dense vector representations (**sentence embeddings**)



Text chunks

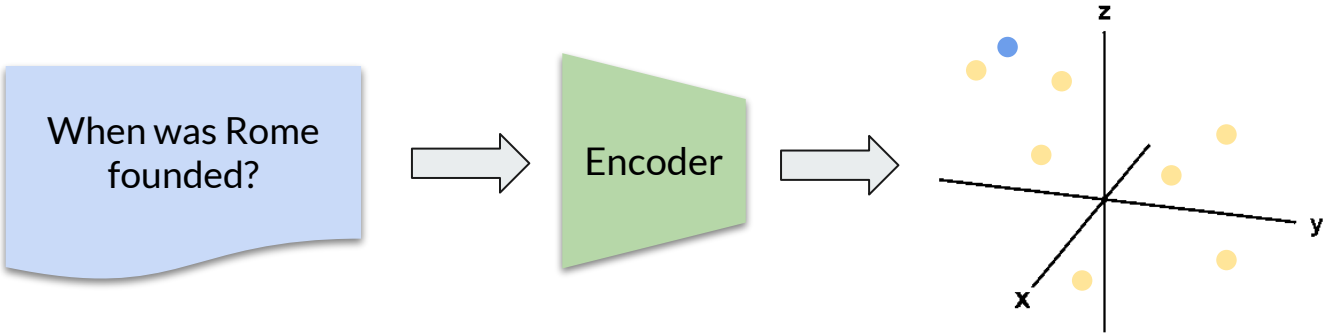
Each point represents a text in the vector space

HW 1 - Task Definition

Retrieval via Semantic Search (3/4)

You will also be given the queries expressing the information need

Goal 2: encode each query into fixed-size, **dense vector representations**, using the same encoder



Query/Question

Encode the question in the same space of the text chunks

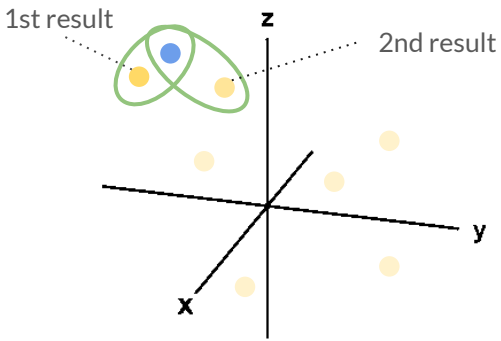
HW 1 - Task Definition

Retrieval via Semantic Search (4/4)

Goal 3: use **similarity metrics** in this embedding space to find chunk candidates closest to the question

Text chunks

Query



$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \, \|\mathbf{b}\|}$$

The most used similarity metric is **cosine similarity**, but any can work

Pick the most similar candidates

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Dataset 🤗

HW 1 - Dataset

Hugging Face Dataset (1/4)

We provide you with a dataset in Hugging Face format, accessible from here.

Load it directly in Python using:

```
from datasets import load_dataset

ds = load_dataset(
    "sapienzanlp-course-materials/hw-mnlp-2026"
)
```

The dataset serves for *both Homework 1 and Homework 2*, but not every feature can be used in Homework 1.

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HW 1 – Dataset

Hugging Face Dataset (2/4)

The dataset has *several features*.

For **Homework 1**, <u>you can **only** use the following features</u>:

Feature	Description
`query (str)`	A single question in natural language
`query_id (str)`	The question (query) identifier
`candidate_chunks (List[str])`	List of candidate answers (only one is correct)
`n_candidates (int)`	Number of candidate answers (length of `candidate_chunks`)
`answer (str)`	The correct answer, an element of `candidate_chunks`
`answer_pos (int)`	Position of `answer` (the correct answer) in `candidate_chunks`

Note: `answer` and `answer_pos` are *not* available in the **blind split**!

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HW 1 - Dataset

Hugging Face Dataset (3/4)

The provided dataset is divided into the following **three splits:**

1. **Train/Dev split** (8,000 examples)
   Used to **fine-tune your models** and experiment with different strategies (you can use a subset of it for validation)
2. **Test split** (2,000 examples)
   Used for **testing your implementation** and comparing your results against baseline models. Must **not** be used for training.
3. **Blind split** (1,322 examples)
   Used for **our final evaluation**. Correct answers are **not** provided

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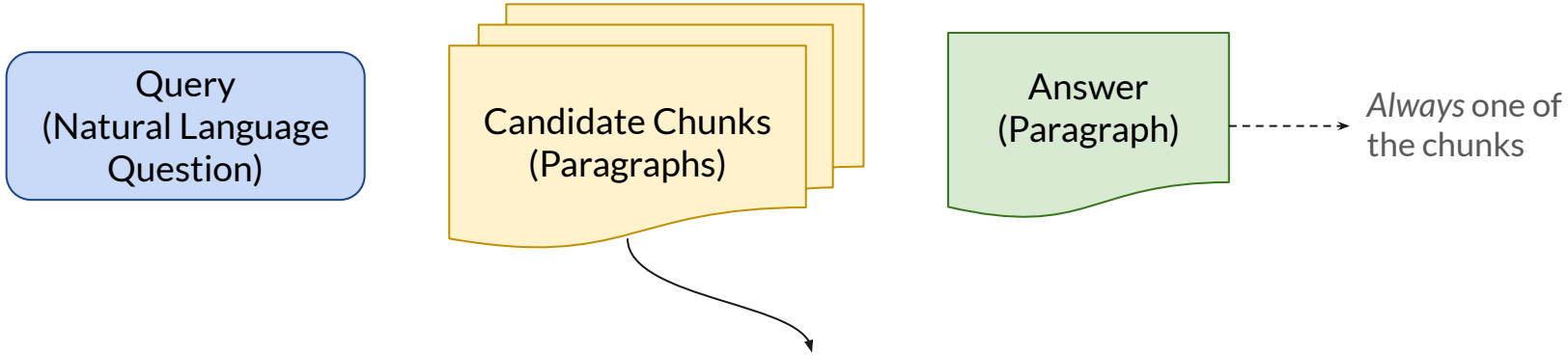
HW 1 - Dataset

Hugging Face Dataset (4/4)

The most **relevant features** of the dataset are:



"Chunks" are arbitrary text splits: we split a full text (e.g., full Wikipedia page) into smaller chunks to encode them in the representation space.

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Implementation Guidelines

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HW 1 - Implementation Guidelines

Overview

There are two types of requirement:

Base requirements

All base requirements must be implemented by *everybody.*

Additional Requirements

Attending students *must* implement *at least one* additional requirement.

Non-attending students *must* implement *at least two* additional requirements.

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HW 1 - Base Requirements

Overview

The following base requirements must be implemented by all students:

- **[B1] Baselines implementation** → Implement the two baseline semantic search systems; the baselines have to match our results
- **[B2] Model fine-tuning** → Fine-tune *at least* distilbert-base-uncased using *at least* one fine-tuning strategy
- **[B3] Written report** → Illustrate your implementation and discuss results on the **test split**. Focus on comparison between baselines and fine-tuned models, trade offs, etc. **Everything you do must be justified and discussed/commented!**
- **[B4] Output files** → Submit your output files for the **test** and **blind splits** in the requested format

Details about **[B1]** *and* **[B2]** *in the next slides. For* **[B3]** and **[B4]**, refer to the "Submission Files" section.

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HW 1 - Base Requirements [B1]

Baselines Implementation – Models

Set up the semantic search system using the following **two baseline models**:

distilbert-base-uncased

A **base encoder model**: it is *not* fine-tuned for sentence-level embeddings.

Sentence-level embeddings can be obtained by averaging the output token-level embeddings (*mean pooling*).

Details about the model here.

all-MiniLM-L6-v2

A **small encoder model** that has **already** been **fine-tuned** for semantic textual similarity tasks.

Therefore, it should already perform well on many retrieval-based tasks.

Details about the model here.

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HW 1 - Base Requirements [B1]

Baselines Implementation – Evaluation Metrics (1/2)

Implement **from scratch** and compute **Hit@k metric** (with **k = [1, 3, 5]**), and report the results on the **test set**. ***No use of LLMs is allowed***

$$Hit@k = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\left[r_i \leq k\right]$$

- Assign **1** if the rank of the correct answer is lower or equal than *k*, **0** otherwise
- Average across the test split

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HW 1 - Base Requirements [B1]

Baselines Implementation – Evaluation Metrics (2/2)

Your results must match these **results** on the **test set** using the **two baselines models** and **cosine similarity** as similarity metric:

distilbert-base-uncased

Hit@1	0.1765
Hit@3	0.4435
Hit@5	0.6045

all-MiniLM-L6-v2

Hit@1	0.4480
Hit@3	0.7165
Hit@5	0.8080

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HW 1 - Base Requirements [B2]

Fine-Tuning (1/3)

Fine-tune *at least* the **distilbert-base-uncased** model with *at least* one fine-tuning strategy. **Fine-tuning strategies** include:

- Sentence Pairs
- Sentence Triplets
- Contrastive InfoNCE style
- Hard Negatives
- Your own idea (big plus!!!)

Take a look at this documentation for help with fine-tuning implementation and strategies!

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HW 1 - Base Requirements [B2]

Fine-Tuning (2/3)

Sentence Pairs

Label: **similar**

I play tennis during summer

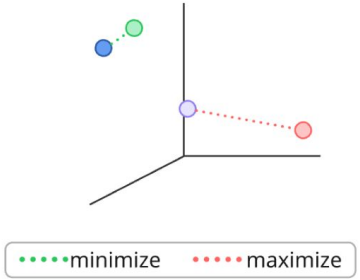
With warm weather, I play tennis

Label: **not similar**

She was doing her homework.

My activity is designing furniture.

Training objective



Sentence Triplets

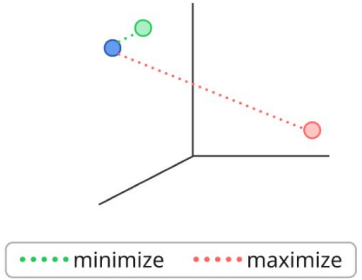
Label: **anchor, positive, negative**

I play tennis during summer

With warm weather, I play tennis

She was doing her homework.

Training objective



InfoNCE Contrastive Loss

Label: **related + *k* unrelated texts**

I play tennis during summer

With warm weather, I play tennis

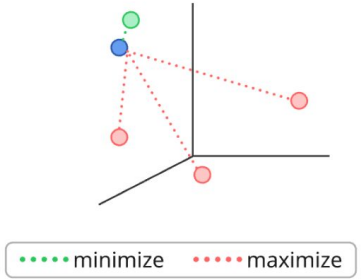
She was doing her homework.

⋮

My activity is designing furniture.

Jack asked where were you.

Training objective



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HW 1 - Base Requirements [B2]

Fine-Tuning (3/3)

You must not include data from either the test split or the blind split when fine-tuning, **only use the train/dev split**.

Each fine-tuned model **should be compared** with the corresponding baseline model.

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HW 1 - Additional Requirements

Overview

Concerning **additional requirements** (this slide), you have the following options:

- **[A1]** (Easy): Report **additional performance metrics** beyond Hit@k
- **[A2]** (Easy): Implement **alternative similarity metrics** and compare them with cosine similarity
- **[A3]** (Easy): Use **additional baseline, off-the-shelf models** and **report a more thorough comparison** with your model(s)
- **[A4]** (Medium): **Fine-tune multiple models**/use **multiple fine-tuning strategies**
- **[A5]** (Hard): Implement a **re-ranking logic** on top of your retrieval one
- **[A6]** (Very Hard): Experiment with **alternative chunking strategies**

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HW 1 - Additional Requirements [A1]

Alternative Similarity Metrics

You can use any **similarity** (or distance) **metrics** that is defined in the vector space.

Note: similarity metrics can be simply (`1 - distance`)

Some of the most famous ones include:

- Euclidean distance
- Angular distance
- ...

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HW 1 - Additional Requirements [A2]

Alternative Similarity Metrics

You can also explore additional retrieval metrics and choose the most suitable ones (search the Web, use LLMs to find alternatives, be creative).

Example: **Mean Reciprocal Rank (MRR)**

$$\mathrm{MRR} = \frac{1}{N} \sum_{q=1}^{N} \frac{1}{\mathrm{rank}_q}$$

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HW 1 - Additional Requirements [A3]

Alternative Similarity Metrics

We strongly encourage you to experiment with **different baseline methods**:

- Especially *off-the-shelf* models already fine-tuned for sentence embeddings
- Chance to **enrich your report** with interesting results and thorough discussions and comparisons

Note: not beating an off-the-shelf model does not mean your homework is bad! Many of these models are trained on billions of texts.

The important thing is that the results are **discussed thoroughly**.

HW 1 - Additional Requirements [A4]

Fine-tune multiple models

Fine-tuning allows the model to capture the specific notion of similarity required for the retrieval task.

Suggestions:

- **Architecture**: You can fine-tune an off-the-shelf model or add your own customizations to the architecture
- **Strategy**: You can experiment with techniques to improve stability, such as early stopping and loss regularization.

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HW 1 - Additional Requirements [A5]

Re-Ranking (1/3)

Re-ranking is the *second stage scoring step* that refines the results returned by the initial retrieval model:

- **Initial retriever**: deals with the entire knowledge base and is optimal for efficiency (not necessarily for precision)
- **Re-ranking:** deals with a subset of candidates and is optimal for precision (not necessarily for efficiency)

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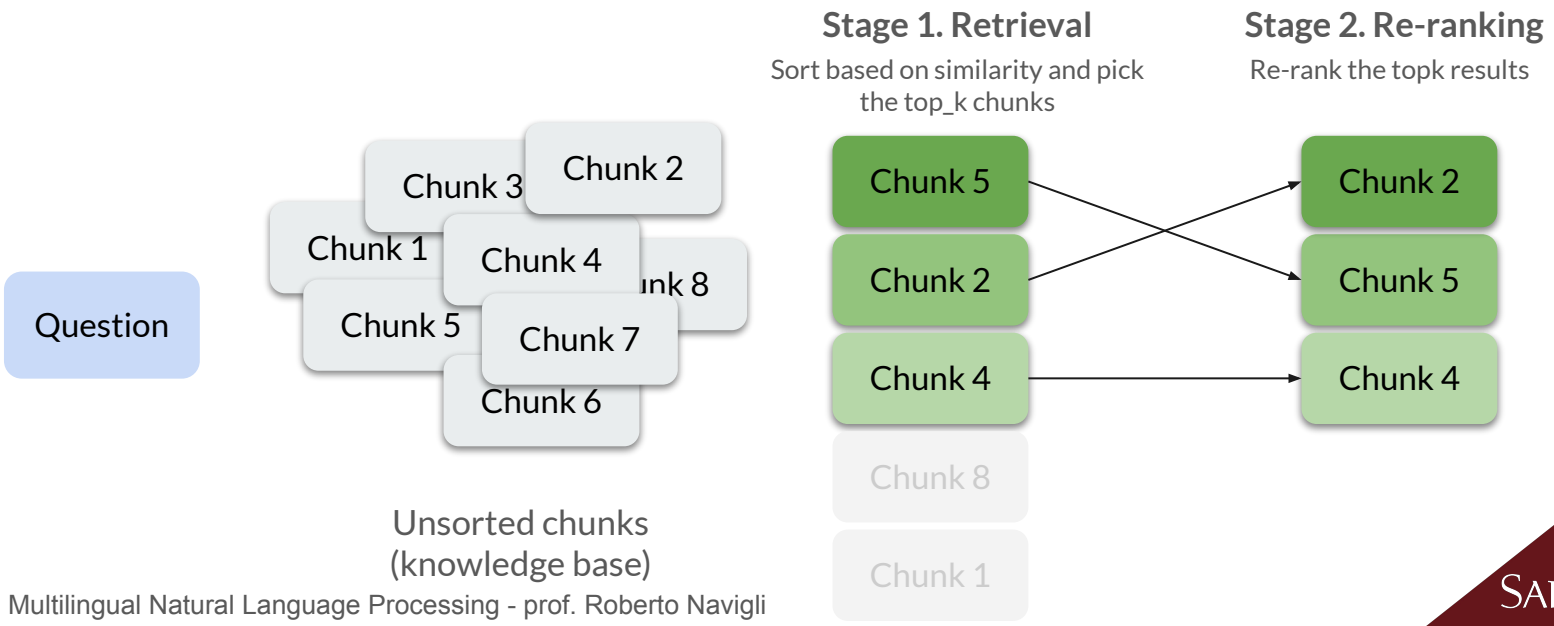
HW 1 - Additional Requirements [A5]

Re-Ranking (2/3)

After sorting the chunks based on their similarity with the question, you can apply a **re-ranking logic to create a new ranking for the *top_k* extracted results**.



HW 1 - Additional Requirements [A5]

Re-Ranking (3/3)

Several ways to implement re-ranking:

- Specialized cross-encoder models (use off-the-shelves)
- Keyword matching methods
- ...

Experiment and discuss the trade-off between performance and efficiency.

If you decide to implement re-ranking, make sure to discuss *both* the standard results and the re-ranked results, and to **provide us with the results of both standard and re-ranking strategies on the blind set as separate JSON files!**

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HW 1 - Additional Requirements [A6]

Alternative Chunking Strategy

You can experiment with **different text-splitting techniques** to create **more informative chunks**. Your final `n_paragraphs` must be **higher or equal** to the original one.

Please note that changing the chunks means no longer having a 1-to-1 correspondence with the answer (which *is* an original chunk).

Thus, you have to:

- Explain your **splitting logic** thoroughly
- How you **map** the original answer chunk to your new chunks
- How you **evaluate edge cases** (e.g., only part of the original answer is retrieved)
    - You might need *additional retrieval metrics*...

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HW 1 - Recap

Base and Additional Requirements (1/3)

Everything in the **base requirements** must be implemented <u>by all teams.</u>

In addition:

- **Attending students** need to choose <u>*at least one*</u> **additional requirement** to implement
- **Non-attending students** need to choose <u>*at least two*</u> **additional requirements** to implement, one of which <u>**must not**</u> be easy (**not** [A1], [A2], or [A3])

Attending students:	Base Requirements	+	≥ 1 Additional Requirements
Non-attending students:	Base Requirements	+	≥ 2 Additional Requirements

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HW 1 - Recap

Base and Additional Requirements (2/3)

For example, an **attending** student might choose:

Base Requirements

- [B1] Baselines implementation
- [B2] Fine-tuning distilbert-base-uncased
- [B3] Report (compare and discuss)
- [B4] Output on test and blind splits

+

Additional Requirements

- [A3] Additional baseline models

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HW 1 - Recap

Base and Additional Requirements (3/3)

For example, a **non attending** student might choose:

Base Requirements

- [B1] Baselines implementation
- [B2] Fine-tuning distilbert-base-uncased
- [B3] Report (compare and discuss)
- [B4] Output on test and blind splits

+

Additional Requirements

- [A3] Additional baseline models
- [A4] Fine-tune an additional model

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Submission Files

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HW 1 - Submission Files

1. **Report**: follow the specific guidelines provided in the slides below

2. **Code**: Colab notebook with runnable code
   a. Well-formatted and well-commented
   b. Everything needs to be reproducible and ready to run

3. **Output files**:
   a. Submit the **JSONL files** for your implementation output(s) on **test** and **blind** sets
   b. Include `README.txt` explaining the contents of each JSONL file

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Report (1/3) [B3]

MAX 2 PAGES for main part of the report
+
No limit Appendix with Images and Table

- The report must be written in **ACL format** and properly structured
- We suggest using Overleaf for report creation
- The report should include the following clearly separated sections:
  - **Introduction** (***VERY VERY VERY*** brief)
  - **Methodology**: justify and explain the methods you used to carry out the homework
  - **Experimental Setup:** explain your experimental setup (*how* you implemented what you described before)
    - **Only** here you give technical details on personalized data/organization/additions, models, baselines, etc. [do not provide setup information that is given by us, but your own choices]
  - **Results**: present the results referencing tables and **commenting** them
    - Highlight the experimental findings with comparison across systems/baselines
    - How you explain the (most notable/unexpected) results

Use the Appendix **only** for figures, tables, pseudo code and additional, specific explanations.

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Report (2/3) [B3]

MAX 2 PAGES for main part of the report
+
No limit Appendix with Images and Table

- Mixing content between sections or presenting information in an unclear or disorganized manner will result in a lower grade (Low-Pass)
- All **images, tables, and results** must be clearly referenced and discussed in the "Results" section.
  - Tables and figures **should not be screenshots** (import images in `.pdf` for better rendering)

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Report (3/3) [B3]

MAX 2 PAGES for main part of the report
+
No limit Appendix with Images and Table

- The **Methodology** should be detailed and clear enough to be understood by a reader with your knowledge of the topic
  - Do not waste space by explaining to us the course core concepts (e.g., Tokenizers, Transformers, ...)
  - Write a concise description if what you're using is not covered in the course and is something relatively new/not commonly used
- **Appendix:**
  - Use the Appendix to further elaborate on your choices, additional details and/or all the combinations you tried / everything worth mentioning that is not in the main body
  - **Note:** the main body of the report shall contain ***ONLY*** the final work you're presenting (i.e., *what worked, not all your trials*)
    - Be **merciless** with what to include, what to put in the appendix, and what to NOT include

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Code [B4]

- The code must be in **Colab notebooks,** runnable, and well formatted
  - If your code does not run, depending on severity you will get a Low-Pass or a Fail **at our own discretion**

- The code should be **commented** and **clear**

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Output Files (1/4) [B4]

- All output files must be placed in the designated **shared folder**
- Ensure that **access permissions** are properly set so that we can view and download all necessary files
- The output files must exactly match the specified format (see slides below)

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Output Files (2/4) [B4]

- Provide **semantic search outputs** on **test** and **blind** splits
- Output files must be a **JSONL file** following the specified format:

```
{query_id_1: [3, 0, 1, 2]}
{query_id_2: [5, 2, 0, 4, 3, 1]}
{query_id_3: [1, 4, 3, 2, 0]}
```

where:

➔ the keys are the `query_id` we provide in the dataset

➔ the list of integers is your ranking of chunks, sorted by your chosen similarity metric (or re-ranker)

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Output Files (3/4) [B4]

If you have **multiple outputs** (e.g., multiple fine-tuned models, different fine-tuning strategies, re-ranking, ...), provide **one JSONL file for each implementation**.

Output Files (4/4) [B4]

Follow this **naming convention**:

- **Test set results:**
  - `[group_name]-test-[model_name]-[variant].jsonl`
- **Blind set results:**
  - `[group_name]-blind-[model_name]-[variant].jsonl`

Please, use reasonable names for your `variant` suffix.

In any case, submit also a `REAMDE.txt` file that tells us **exactly what each JSONL file represents**.

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Submission guidelines

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Submission procedure - VERY IMPORTANT!!

⚠️

1. **12th April:** Complete the **group form** uploaded in Classroom so we know how many of you are submitting
2. **19th April:** Submit using the **submission form** (***NOT* the above form**)
   a. You have to submit a Google Drive folder. **The folder must be shared with us (e-mail specified on slide 10).**

ONLY ASSIGNMENTS SUBMITTED THROUGH THE SUBMISSION FORM WILL BE ACCEPTED!!!!!!!!!

If you share your homework with us in any other way, **we will not accept it.**

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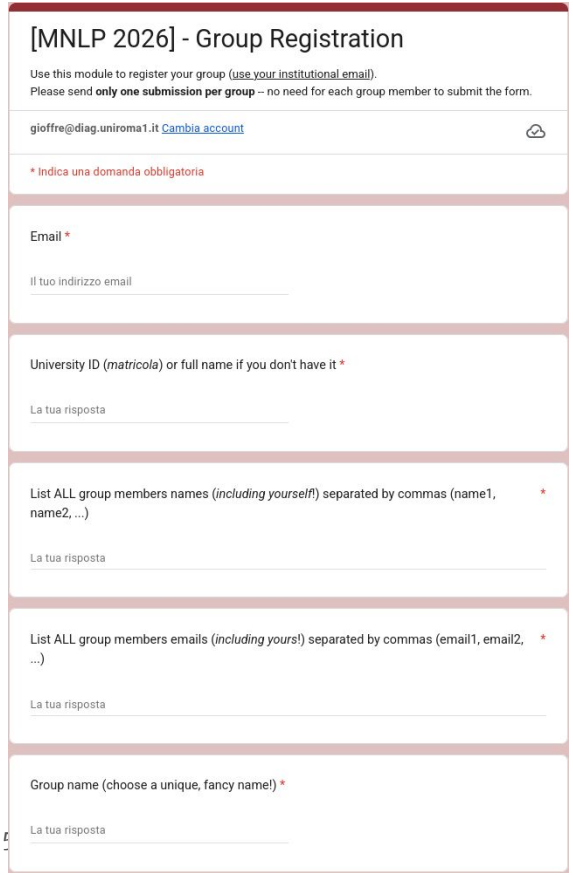
Submission instructions



[MNLP 2026] - Group Registration

Use this module to register your group (use your institutional email).
Please send **only one submission per group** -- no need for each group member to submit the form.

gioffre@diag.uniroma1.it Cambia account

* Indica una domanda obbligatoria

Email *

Il tuo indirizzo email

University ID (*matricola*) or full name if you don't have it *

La tua risposta

List ALL group members names (*including yourself!*) separated by commas (name1, name2, ...) *

La tua risposta

List ALL group members emails (*including yours!*) separated by commas (email1, email2, ...) *

La tua risposta

Group name (choose a unique, fancy name!) *

La tua risposta

When submitting using the form:

- Use your institutional email (**only one email - and always the same email - represents the group**)
- Please write your group members with the same spelling
- *Do not abuse* the additional notes field (one short line if needed, otherwise send us an email)
- **Double check** that the files you are sharing are accessible from us!

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Plagiarism

We will check for plagiarism both manually and automatically.

It is **not allowed** to:

- Copy code or reports from other students/groups
- Share your code with other students outside your group
- Use LLMs to write code and the report

Projects violating any of the above conditions will be **desk-rejected**

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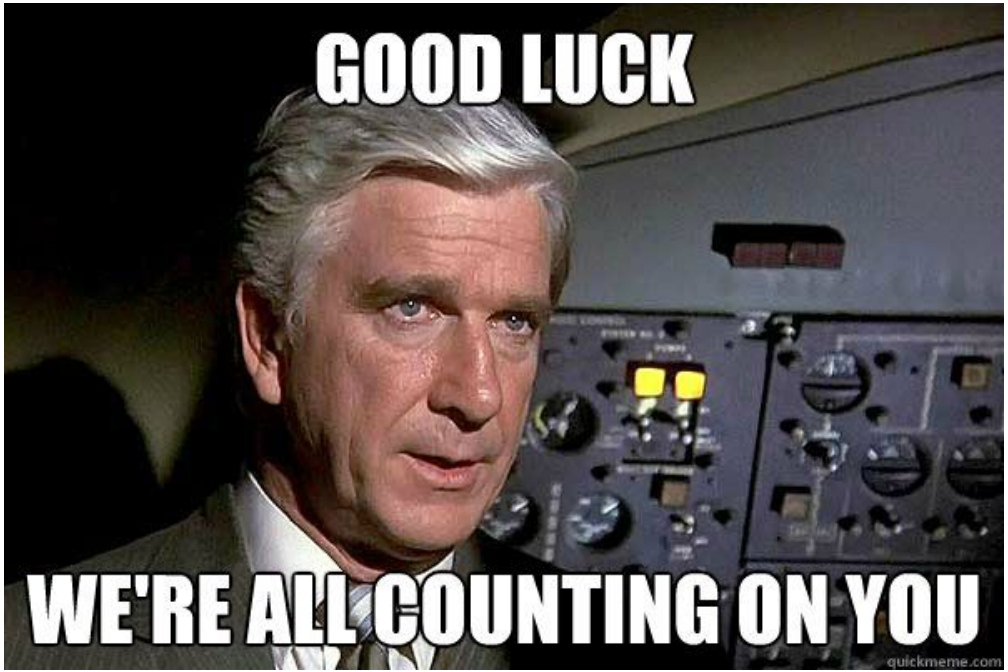

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